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Normal Vector Estimation in 3D Point Clouds

Internship Report

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Abstract

Normal vector estimation plays a fundamental role in 3D computer vision and geometric data processing. This report presents the methodology and outcomes of a summer internship conducted at CDTA, focusing on implementing and evaluating a normal vector estimation technique for 3D point clouds using Principal Component Analysis (PCA) within a **Planar Segmentation** framework. The project aimed to design an efficient pipeline for estimating surface orientations, evaluate it on synthetic and real-world data using a quantitative **Planarity Score**, and deploy an interactive visualization platform for demonstration. Through this work, the intern gained experience in 3D geometry, data analysis, and computational design, contributing to the understanding of how classical geometric methods perform under real-world constraints.

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1 Introduction

With the rapid advancement of 3D sensing technologies such as LiDAR, stereo vision, and structured light, 3D point clouds have become a major source of spatial information in modern applications — from autonomous navigation to digital twins and augmented reality. However, raw point clouds only provide coordinates of sampled points without any notion of surface continuity or orientation. To interpret such data, it is essential to estimate local geometric properties — primarily, **surface normals**.

Surface normals describe the direction perpendicular to a local surface patch and are fundamental for numerous downstream tasks:

- **Surface reconstruction:** estimating continuous surfaces from discrete points;
- **Segmentation and recognition:** distinguishing planar or curved structures;
- **Shading and rendering:** computing realistic lighting and reflections;
- **Registration:** aligning two point clouds by matching their geometric features.

Despite its simplicity in theory, estimating normals in practice is challenging due to:

- sensor noise and outliers,
- irregular sampling densities,
- complex geometries with edges and corners.

This internship focused on developing a robust and interpretable approach for normal estimation using a PCA-based **planar segmentation** approach. The method leverages the statistical properties of point neighborhoods to infer surface orientations and was later integrated into a web application for interactive testing.

1.1 Objectives

The main objectives of this internship were:

1. To conduct a literature review on classical and deep learning methods for normal estimation;

2. To implement a PCA-based approach for 3D point clouds using a **Seeding and Exploration** pipeline;

3. To evaluate the effect of neighborhood parameters and different seeding algorithms on data quality;

4. To develop a simple visualization and deployment tool to demonstrate the results.

1.2 Structure of the Report

This report is organized as follows: Chapter 2 presents the theoretical background of normal vector estimation and PCA. Chapter 3 details the implemented methodology and algorithms. Chapter 4 discusses the obtained results and performance analysis. Finally, Chapter 5 concludes with reflections and future directions.

2 Theoretical Background

Normal estimation can be viewed as a local geometric approximation problem. Given a set of 3D points, the goal is to find, for each point, the direction perpendicular to the best-fitting plane passing through its local neighborhood.

2.1 Neighborhood Selection

To compute meaningful normals, we first need to define a local region around each point. Two common neighborhood definitions are:

- **k-Nearest Neighbors (kNN):** select the k closest points to a target point based on Euclidean distance;
- **Radius Search:** select all points within a fixed radius r.

Both have trade-offs: kNN ensures constant computational cost, while radius search adapts better to variable densities.

2.2 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a statistical tool used to identify the directions in which data varies the most. For a local neighborhood $\{p_1, p_2, ..., p_n\}$, the mean point (centroid) is computed as:

$$\bar{p} = \frac{1}{n} \sum_{i=1}^{n} p_i$$

The covariance matrix is then given by:

$$C = \frac{1}{n} \sum_{i=1}^{n} (p_i - \bar{p})(p_i - \bar{p})^T$$

The eigenvectors of C represent orthogonal directions (axes) that describe how the points are distributed in 3D space, while the eigenvalues measure how much variance exists along each axis.

Since a local surface is approximately planar, most variation occurs along the tangent plane. Thus:

- The two largest eigenvalues correspond to directions in the tangent plane;
- The **smallest eigenvalue** (λ_3) corresponds to the direction with minimal variation — the **surface normal**.

This geometric intuition makes PCA particularly effective for estimating normals from noisy point data.

2.3 Geometric Feature Estimation

The three eigenvalues ($\lambda_1 \geq \lambda_2 \geq \lambda_3$) of the covariance matrix describe the local geometry: λ_3 corresponds to the normal direction, while λ_1 and λ_2 span the tangent plane. These values are used to quantify the local shape, crucially, the **Planarity Score**:

$$\text{Planarity} = \frac{\lambda_2 - \lambda_3}{\lambda_1}$$

A value closer to 1 indicates a strong planar structure, which is essential for filtering initial seed points and evaluating the quality of segmented regions.

2.4 Normal Orientation

Because PCA only defines the direction of the normal, not its orientation, normals across the surface may be inconsistently flipped. To correct this, orientation consistency is enforced using a propagation method — aligning each normal to face roughly the same direction as its neighbors or relative to a known viewpoint.

3 Methodology

The normal estimation and planar segmentation pipeline implemented during the internship consists of the following main stages:

1. Point cloud acquisition or loading;
2. Neighborhood computation;
3. **Planar Segmentation** via Seeding and Exploration;
4. Normal orientation correction;
5. Visualization and evaluation.

3.1 Implementation Environment

The project was implemented in **Python 3.10** using the following libraries:

- **Open3D:** for point cloud processing and visualization;
- **NumPy:** for numerical operations;
- **Matplotlib:** for plotting performance metrics;
- **Flask:** for web-based deployment.

3.2 Algorithmic Steps

The robust normal vector estimation was achieved through a two-stage **Planar Segmentation** pipeline, integrating PCA for local analysis:

1. **Seeding Algorithm:** For each unsegmented point p_i, identify its k-nearest neighbors and compute the Planarity score. Select an initial seed point p_{seed} that exhibits the highest local planarity and satisfies a threshold on the smallest eigenvalue $\lambda_3 < \epsilon$.

2. **PCA-based Normal Estimation:** Compute the local covariance matrix for the seed point neighborhood and perform eigen-decomposition to find the normal vector (eigenvector corresponding to λ_3).

3. **Exploration/Region Growing Algorithm:** Iteratively add neighboring points to the planar region if their computed normal is within a specified angular threshold of the seed's normal, effectively growing the planar segment.

4. **Normal Orientation Correction:** Reorient all final normals within the segment for global consistency relative to a known viewpoint.

3.3 Parameter Study

Beyond the neighborhood size k, a comparative study was conducted on different methods for selecting the optimal **initial seed point** for segmentation, which significantly impacts performance:

- **Regular k-Nearest Neighbors (kNN):** Neighborhood based purely on proximity.

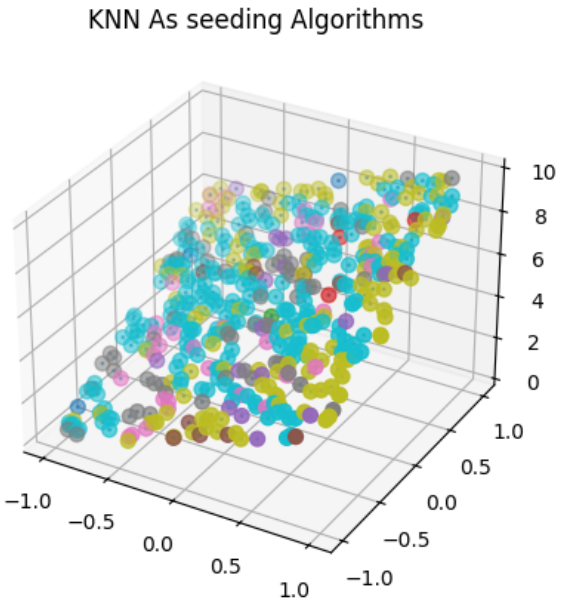


Figure 3.1: The Seeding possible points resulting from using KNN.

- **Weighted kNN:** Neighbors weighted inversely by distance.

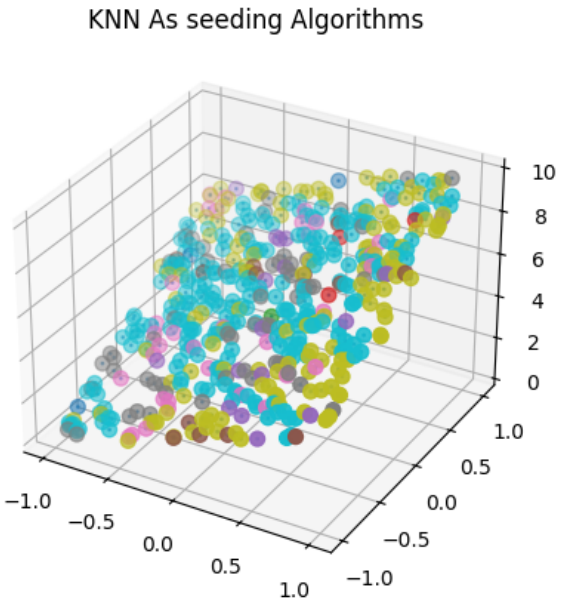


Figure 3.2: The Seeding possible points resulting from using weighted KNN.

- **Highest Local Planarity:** Selecting the point whose neighborhood yields the highest Planarity Score, indicating the best local fit to a plane.

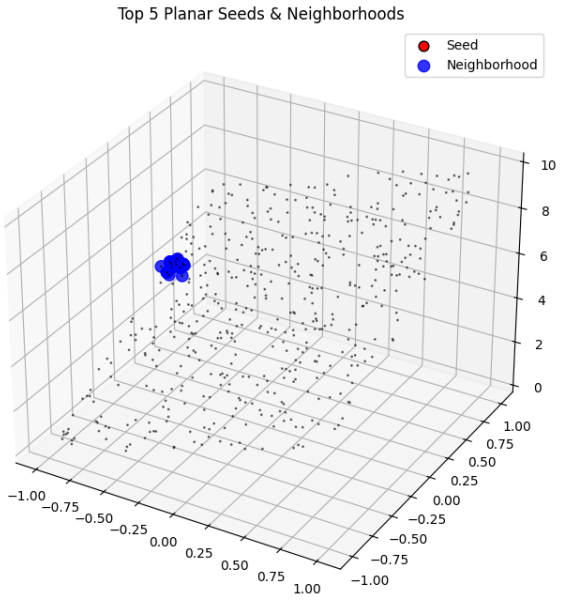


Figure 3.3: The Highest planarity point resulting from using the highest planarity.

The planar-based seeding method was found to be the most robust. Typical values for k (number of neighbors) ranged from $k = 10$ to $k = 50$.

4 Results and Discussion

4.1 Visualization

Figure 4.5 shows an example of estimated normals on a planar surface. Each arrow indicates the computed normal direction at the corresponding point, demonstrating that PCA correctly recovers perpendicular orientations to the plane.

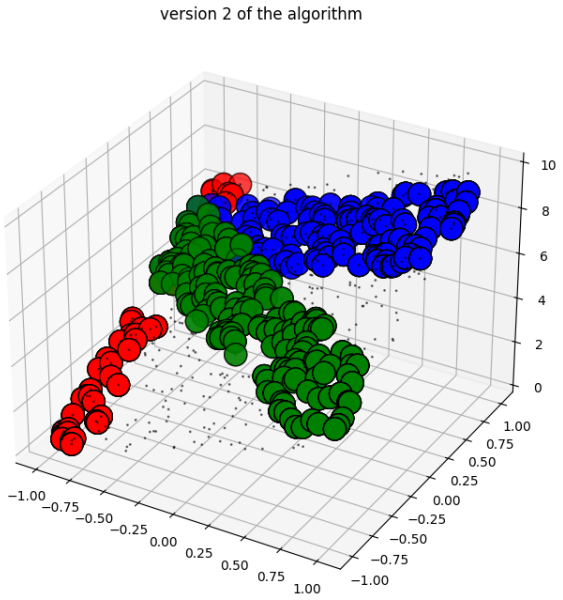


Figure 4.1: Example of PCA-based normal estimation on a planar region using an initial version of seeding.

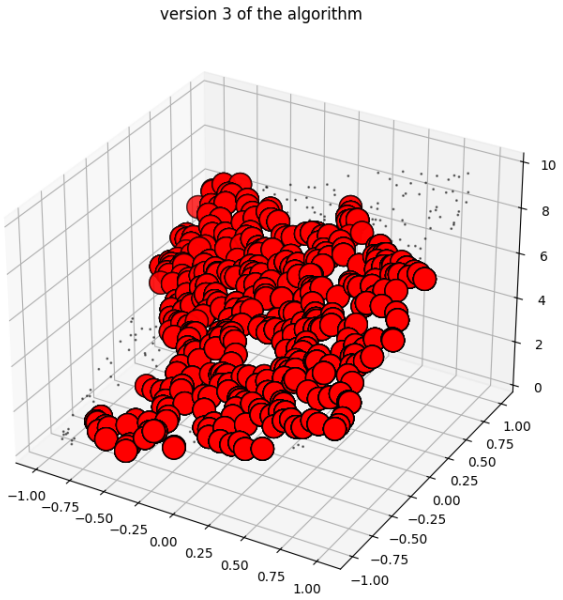


Figure 4.2: Example of PCA-based normal estimation on a planar region using a planarity seeding.

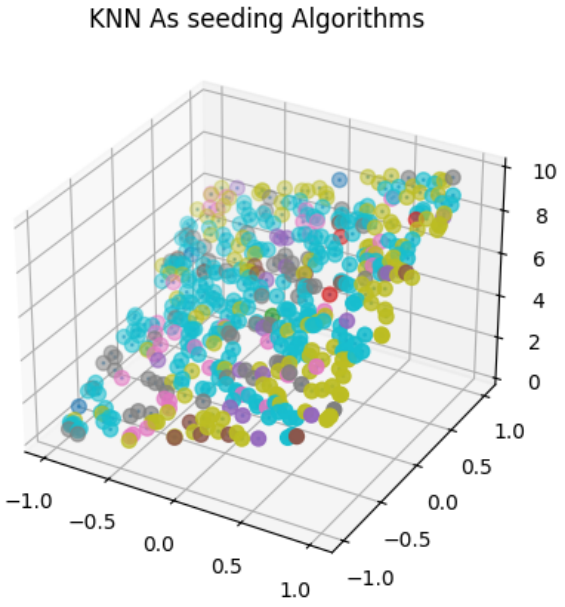


Figure 4.3: The Seeding possible points resulting from using weighted KNN.

We can clearly see from the graphs above, that the algorithm went through different iterations where we tried to improve the performance against edge cases that reflects real-world cases. We even took things further testing the final algorithm against the following edge case.

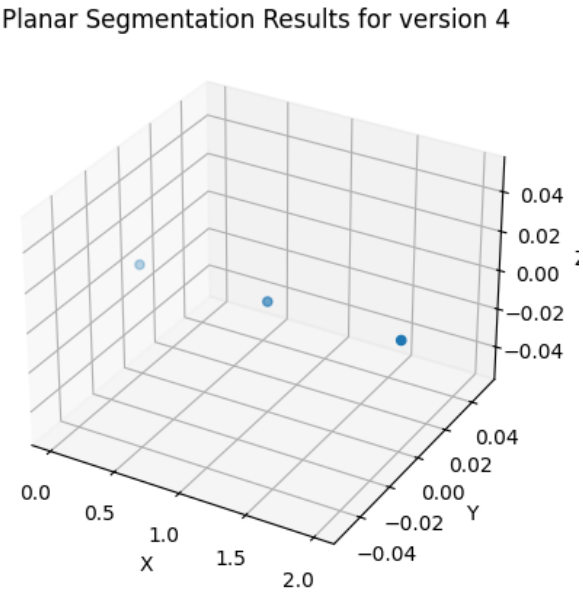


Figure 4.4: Testing the 4th algorithm against degenerate case

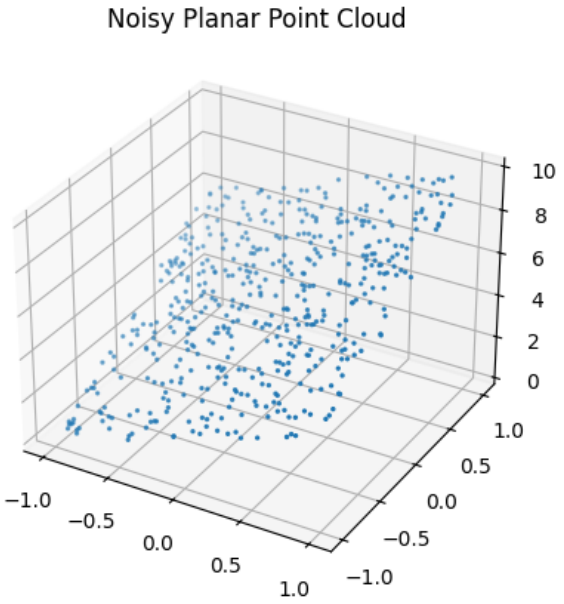


Figure 4.5: The results of the algorithm against the edge case of noisy points

4.2 Parameter Influence

A quantitative comparison revealed that increasing k:

- reduces noise sensitivity;
- smooths local variations;
- but may blur fine surface details.

4.3 Evaluation on Synthetic and Real Data

On synthetic datasets (e.g., cubes and spheres), PCA estimated normals accurately, with angular deviations below 2° for $k \geq 30$. The segmentation quality was quantitatively tracked using the defined **Planarity Score** (Planarity $= \frac{\lambda_2 - \lambda_3}{\lambda_1}$), ensuring that detected segments exhibited high-confidence planar structures (score > 0.8). For real scans, additional preprocessing (e.g., voxel downsampling) improved stability.

4.4 Limitations

While PCA provides interpretable and efficient estimation, its limitations include:

- Poor performance in highly curved or noisy areas;
- Sensitivity to uneven sampling;
- Dependence on proper neighborhood size.

5 Conclusion and Future Work

This internship provided valuable hands-on experience in the geometric analysis of 3D data. Through the study and implementation of PCA-based planar segmentation and normal estimation, it became clear that traditional linear methods still serve as reliable and computationally efficient tools for understanding surface structure.

Future work may focus on:

- Extending to noise-robust or adaptive PCA variants;
- Integrating region-growing or machine learning–based refinement;
- Applying the results to real 3D reconstruction or robotic perception pipelines.

Overall, this project strengthened both theoretical understanding and technical practice in computational geometry and 3D data processing.

References

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